Nonlinear Multilevel Model for Weekly Sales Prediction

Abstract

The primary objective of this study is to analyze the factors influencing Walmart's sales using Machine Learning under the statistical learning paradigm with a statistical focus, particularly emphasizing the efficiency of Multilevel Modeling (HLM3) "3-Level Hierarchical Linear Model." These concepts were employed to develop a nonlinear hierarchical model with 3 levels. Multilevel models were developed using different distributions (both linear and nonlinear) and compared step by step with the respective non-multilevel models, referred to as "Generalized Linear Models" (GLM), meaning models without random effects. The effectiveness of the multilevel modeling was assessed through the development of null models to diagnose whether the model is more effective when the data is distributed in hierarchical layers. Null models of Linear Regression, Nonlinear Regression using the "Yeo-Johnson" transformation, Linear HLM3 Multilevel Model, and complete models of Linear and Nonlinear Regression were estimated for comparison purposes. Statistical measures such as Log-Likelihood [LogLik], Akaike's Information Criterion [AIC], and Bayesian Information Criterion [BIC] were used to evaluate the performance of each model.
The analysis of the factors influencing Walmart's sales using Machine Learning under the statistical learning paradigm can provide valuable insights for improving sales strategies and decision-making processes. The efficiency of multilevel modeling seems to be a promising approach to handle data distributed across hierarchical layers. The findings of this study contribute to a better understanding of the statistical aspects related to the topic and provide a solid foundation for future research in the field.

Keywords: Yeo-Johnson; Multilevel Model; Hierarchical Model; Machine Learning.

Introduction

The Multilevel Regression Model for Panel Data is derived from regression models for panel data and, according to Fávero and Belfiore (2017), is particularly useful when studying the behavior of a given phenomenon—represented by the dependent variable—in the presence of grouped data structures with repeated or longitudinal measures.

Also according to Fávero and Belfiore (2017), the Multilevel Regression Model for Panel Data has gained considerable importance in various fields of knowledge. The publication of studies has significantly increased, driven largely by the determination of research constructs that consider the existence of nested data structures, where certain variables exhibit variation between distinct units representing groups but not between observations within the same group.

The choice of the Hierarchical Multilevel Model is due to its efficiency in handling hierarchical data and its ability to capture the variation provided by different levels of data grouping, as it captures random effects instead of merely fixed effects. When levels,

heteroscedasticity, non-normality in the residuals even after corrective transformations, and the use of other models that might seem more appropriate but turn out not to be are observed, this model is a strong option to consider. Since the model is nonlinear and there are three levels, one of which is time measured in weeks, the 3-Level Hierarchical Linear Model (HLM3) with Repeated Measures was employed after transformations to stabilize variance and bring the data closer to a normal distribution. As discussed by Fávero, Santos, and Serra (2018a), a researcher can also propose nonlinear hierarchical models.

The proposed modeling framework aims to perform supervised analyses to understand the relationship between the explanatory variables and the dependent variable, comprehend the relationship between the model's levels, assess how each level influences the prediction, and identify the coefficients of the final model equation, thus aiding in the decision-making process.

Although multilevel modeling is an effective approach, this method has some limitations. Among these is the need for an adequate amount of data to ensure robust results, as well as a greater computational processing capacity.

The modeling pipeline adopted in this study followed the stages of predictive model development within the field of Data Science, which consists of:

i) Data collection

ii) Data cleaning

iii) Exploratory data analysis

iv) Feature selection

v) Model creation

vi) Model training

vii) Model evaluation

viii) Model adjustment

Using weekly sales as the dependent variable in the model is effective for analyzing seasonal events such as holidays, Black Friday, etc. In the case of holidays, the week becomes a crucial reference since sales tend to significantly increase about one week before, and after the holiday, stores often begin a season of major discounts. It is clear that inventory management, promotion planning, and resource allocation also benefit from this weekly approach.

Since this study focused on purchases in physical stores, predicting weekly sales becomes more accurate because not all stores open during holidays, and in some locations, none may open at all.

Modeling Framework and Data Preparation

The dataset used in this study was generated from a list of spreadsheets containing data on the weekly sales of products in various Walmart stores in the United States. Some previous studies using this dataset did not use weekly sales per department as the dependent variable, since the department column was removed, and the sales were aggregated by store. This presented a challenge for implementing the model in this study, as the department field is necessary for the HLM3 Multilevel Model.

A key challenge during the data collection and cleaning phase was selecting one or more columns related to discount markdowns, named MarkDown 1, 2, 3, 4, and 5. These represent the markdowns or discounts offered by the stores. Figure 1 presents a graph with all the discounts. There were many missing values (NAs) in MarkDown columns 2, 3, and 4, while MarkDown 1 had only a small number of missing values, and MarkDown 5 had no missing values. Therefore, MarkDown 1 and 5 were kept, as they represent the markdown values at the start and end of the week, respectively. Additionally, the markdown data were only available from November 2011 onward, which led to the selection of the year 2012 for the analysis.

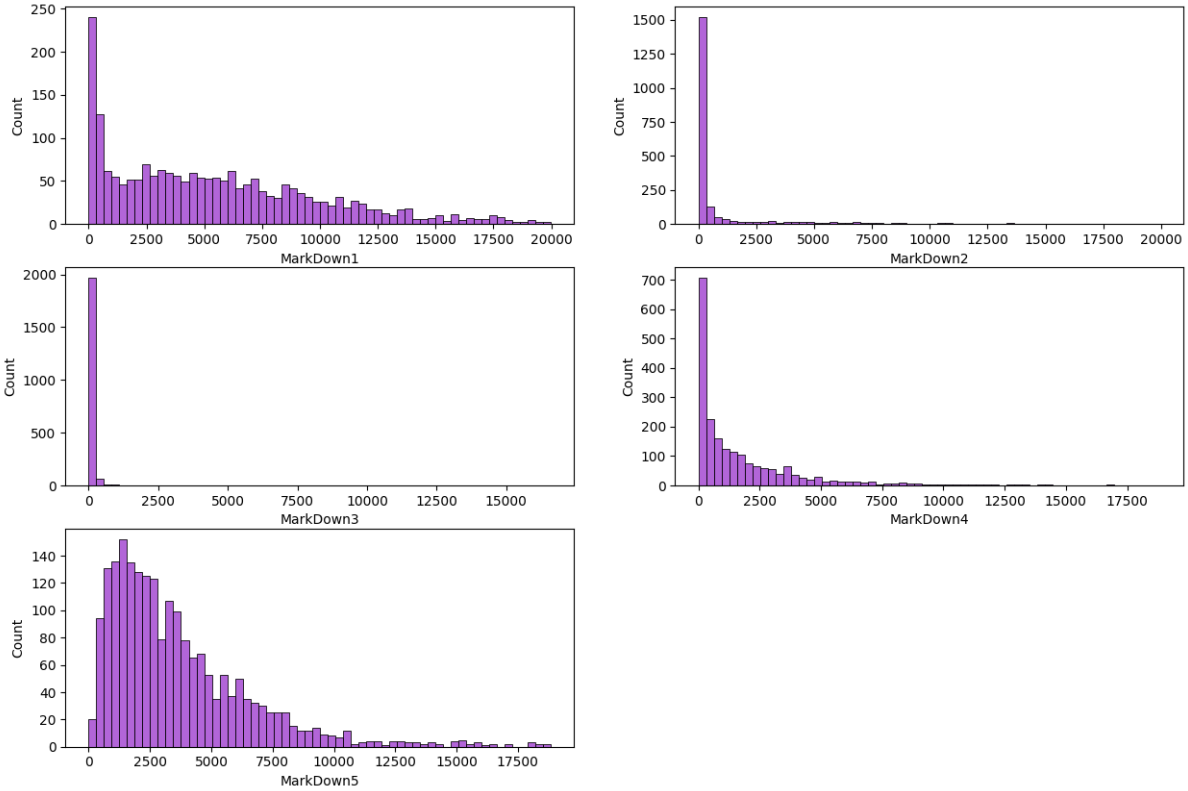

Figure 1. Frequency distributions of the five markdown variables

Below are the descriptions of the columns contained in the dataset:

i) Store: Store code, categorical (1 to 45)

ii) Week: Week number, integer (1 to 42)

iii) Temperature: Temperature, decimal

iv) Fuel_Price: Gasoline price, decimal

v) MarkDown1: Opening discount markdown, decimal

vi) MarkDown5: Closing discount markdown, decimal

vii) CPI: Consumer Price Index, decimal

viii) Unemployment: Unemployment rate, decimal

ix) IsHoliday: Boolean indicating whether it’s a holiday or not

x) Dept: Department code, categorical

xi) Weekly_Sales: Weekly sales, integer

xii) Type: Store type, categorical (A, B, C)

xiii) Size: Store size, integer

The data acquisition, cleaning, and exploratory data analysis were performed using the Python and R programming languages, utilizing Jupyter Notebook and RStudio as the development environments. The model creation, training, evaluation, and adjustment were executed using the R language within the RStudio environment. All of these tools are open-source and free software.

The dataset was imported into the development environment after data cleaning and preprocessing. Table 1 displays the properties of the column values post-cleaning and treatment.

Table 1. Dataset variables, non-null counts, and data types

Column	Non-Null	Dtype
Store	151432	category
Week	151432	int64
Temperature	151432	float64
Fuel_Price	151432	float64
MarkDown1	151432	float64
MarkDown5	151432	float64
CPI	151432	float64
IsHoliday	151432	category
Dept	151432	category
Weekly_Sales	151432	int64
Type	151432	category
Size	151432	int64
Unemployment	151432	float64

Statistical data for each numerical column before applying the Z-Score technique and removing outliers are presented in Tables 2 and 3.

Table 2. Descriptive statistics of Temperature, Fuel_Price, MarkDown1, and MarkDown5

Estatísticas	Temperature	Fuel_Price	MarkDown1	MarkDown5
Count	151432	151432	151432	151432
Mean	15.4445	3.6587	7210.48	4628.97
Min	13.6333	3.031	0	135.16
25%	8.4	3.476	2211.62	1878.44
50%	16.0444	3.666	5320.71	3359.45
75%	23.2611	3.864	9198.08	5563.80
Max	37.8167	4.468	88646.80	108519
STD	9.801998	0.289306	8286.287331	5962.883796

Table 3. Descriptive statistics of CPI, Unemployment, Weekly_Sales, and Size

Estatísticas	CPI	Unemployment	Weekly_Sales	Size
Count	151432	151432	151432	151432
Mean	0.174780	7.401	16174.29	136361.49
Min	0.129817	3.879	0	34875
25%	0.136491	6.334	1997.75	93638
50%	0.189062	7.244	7537.50	140167
75%	0.219905	8.256	204.16	202505
Max	0.227233	12.890	470000	219622
STD	0.040344	1.723361	23178.224555	61114.170844

Following this, a procedure was conducted to include categorical variables in the modeling. All categorical variable values were transformed into columns with 0 or 1 values, indicating the presence or absence of that specific category. This method was applied to the `Type` and `IsHoliday` columns. In non-multilevel models, the same procedure was also applied to the `Store` column.

The technique of assigning numerical values to represent each value in categorical columns is not recommended as it often leads to problems and distortions in the final model. For example, replacing the values “house,” “apartment,” and “farm” in the “domicile” column with 1, 2, and 3, respectively, would introduce bias.

In the frequency graph of the dependent variable (Figure 2), it was observed that the distribution resembled a Negative Binomial Poisson distribution. However, the values were decimals, and some were negative, which is not permitted when applying a Negative Binomial Model. Therefore, negative values associated with returns were not modeled in this case and were treated as a net absence of sales, and decimal values were converted to integers.

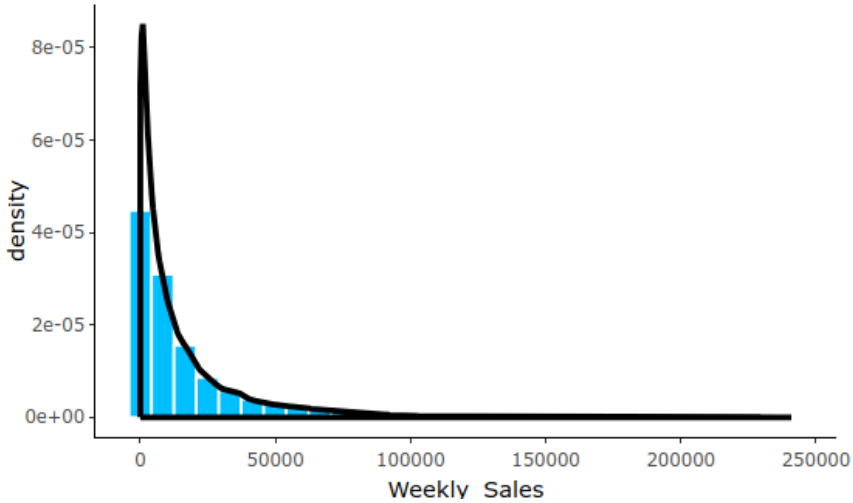

Figure 2. Distribution of the dependent variable Weekly_Sales

One of the conditions for successfully implementing multilevel modeling is that no lower-level index can belong to more than two groups at the upper level. For example, no neighborhood can belong to two different cities. In this study, the second level is the department, and each department is numbered from 1 up to the total number of departments within a store, which is the third level. Therefore, It was required to adjust the department indices for each store. For example, Store 1 has 90 departments, numbered between 1 and 90. The next store has 80 departments, initially numbered between 1 and 80. Following the multilevel modeling rule, the department indices in Store 2 were adjusted to the range of 91 to 170, and so on.

Finally, the numerical data was standardized to ensure that they were on the same measurement scale. The data standardization was performed using the Z-Score technique. Outliers were handled using the Interquartile Range (IQR) method, as some values in the “Weekly_Sales” column were excessively high, as shown in Figure 3.

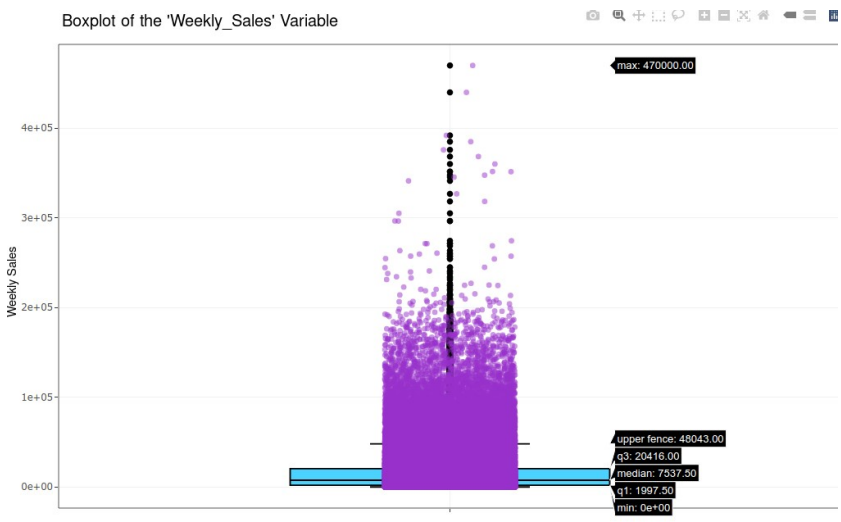

Figure 3. Outlier distribution of the dependent variable Weekly_Sales

It was necessary to verify the presence of multicollinearity, as it could distort the model. Multicollinearity is detected through the calculation of two values: the Variance Inflation Factor (VIF) and Tolerance. VIF is the inverse of the Tolerance value, and both metrics diagnose multicollinearity. If the Tolerance value is close to zero, it indicates high multicollinearity, whereas values close to 1 indicate low multicollinearity. Similarly, if the VIF tends toward infinity, it indicates high multicollinearity, while values close to 1 indicate low multicollinearity. In this thesis, no cases of multicollinearity were detected, as shown in Table 4.

Table 4. Variance Inflation Factor (VIF) and Tolerance values

Variáveis explicativas	VIF	Tolerance
const	6.097479	0.164002
Week	1.823292	0.548459
Temperature	1.745910	0.572767
Fuel_Price	1.452875	0.688290
MarkDown1	1.188364	0.841493
MarkDown5	1.136908	0.879578
CPI	1.344275	0.743895
Unemployment	1.238357	0.807522
Size	1.247109	0.801854

As the objective of this study is inferential and comparative across modeling structures, rather than focused on out-of-sample predictive optimization, the models were estimated using the full dataset. Evaluation was conducted using likelihood-based criteria (LogLik, AIC, and BIC), in accordance with the classical statistical paradigm. It is acknowledged that, within the predictive Machine Learning context, a train–test split is widely recommended.

GLM models are referred to as “single equation models.” Multilevel modeling is called a “multiple equations model,” as in addition to the typical terms of the final equation in supervised models, it is also possible to capture the so-called “random effects,” which are interactions between levels and within each level. In the case of a 3-Level Hierarchical Model with Repeated Measures HLM3, there are 7 types of random effects. The random effect of level 1 "Time" is called Idiosyncratic Error [ε], the intercept effect [v0jk] of level 2, the slope effect [v1jk] of level 2, the intercept effect [τ00k] of level 3, and the slope effect [τ10k] of level 3. The level 2 effects in this study mean “Are there differences in weekly sales between departments within the same store over time in terms of intercept and slope (or growth rates)?” and the level 3 effects in this study mean “Are there differences in weekly sales between departments of different stores over time in terms of intercept and slope?”. The indices t, j, and k are respectively the indices of time, department, and store.

Figure 4 shows the structure of the HLM3 multilevel model.

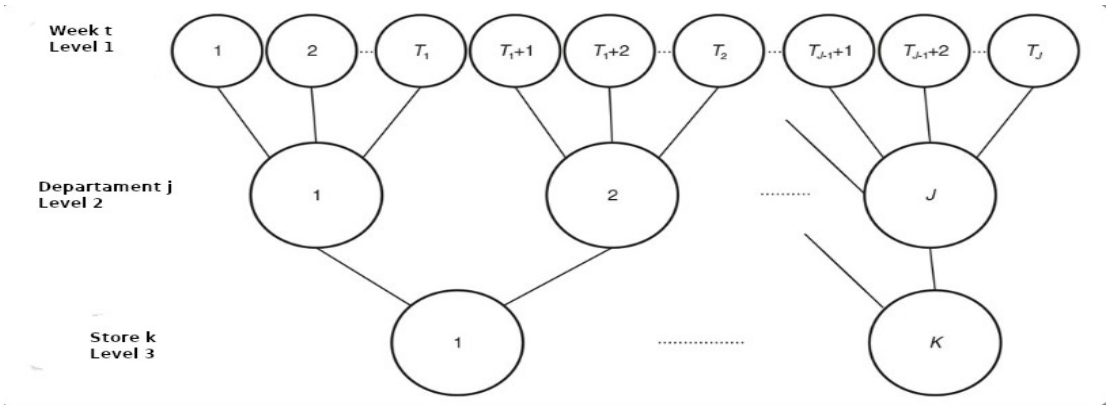

Figure 4. Three-level hierarchical model with time at level 1
Source: Adapted from Fávero (2024).

In statistical models, when GLM models cannot fit the data well, the model is in well-defined hierarchical levels, and heteroscedasticity in the data is detected, the multilevel model becomes an excellent option. Through tests and graphs, it is possible to verify the presence

of heteroscedasticity. Heteroscedasticity indicates that the dispersion of errors is not uniform and increases as the involved variables increase.

Upon determining that the multilevel model is the best option, it is necessary to build and execute its respective null model. In the null model of a "Machine Learning" model, the dependent variable has no explanatory variables, only the intercept. Its use is frequent in the world of statistical paradigm models (such as GLM and GLLM) to choose the most efficient model by observing statistical metrics such as LogLik, AIC, and BIC. The null model is also used to assess whether the inclusion or exclusion of a particular explanatory variable was beneficial for the model. The best model exhibits a higher LogLik (i.e., less negative) and lower AIC and BIC values.

The multilevel model is built step by step, starting from the null model, according to the "step-up" strategy proposed by Raudenbush and Bryk (2002), as it is not possible to apply the "Stepwise" procedure in multilevel models.

In the case of multilevel modeling, by observing the random values at each level of the null model, it is possible to calculate the influence of each layer on the dependent variable through a statistical measure called the "Intraclass Correlation Coefficient" [ICC]. In this study, the influence of the department, store, and time levels on the dependent variable Weekly_Sales (weekly sales) was calculated.

Equations 1 and 2 were used to develop a nonlinear hierarchical model with 3 levels using the "Yeo-Johnson" transformation

$$\frac{Y_i^{\lambda}-1}{\lambda}=\alpha+\beta_1 . X_{1i}+\beta_2 . X_{2i}+...+\beta_k . X_{ki}+u \tag{1}$$

$$Y_{tjk}=\beta_{0jk}+\beta_{1jk} . X_{jk}+...+\beta_{tjk} . X_{jk}+\varepsilon_{tjk} \tag{2}$$

where Y represents the dependent variable; α: intercept; β: coefficients; X: explanatory variables; k: store index; u: error term; λ: Yeo-Johnson's lambda; t: time index; j: departament index; and ν, τ, and ε represent the random values for the Department, Store, and Time levels, respectively.

The 'Yeo-Johnson' transformation is similar to the "Box-Cox" transformation but, according to Riani (2023), it has the advantage of functioning when the data includes negative numbers and zeros.

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Results and Discussion

Next, It was required to obtain the correlation between the explanatory variables. The correlations were observed through the correlation map in Figure 5.

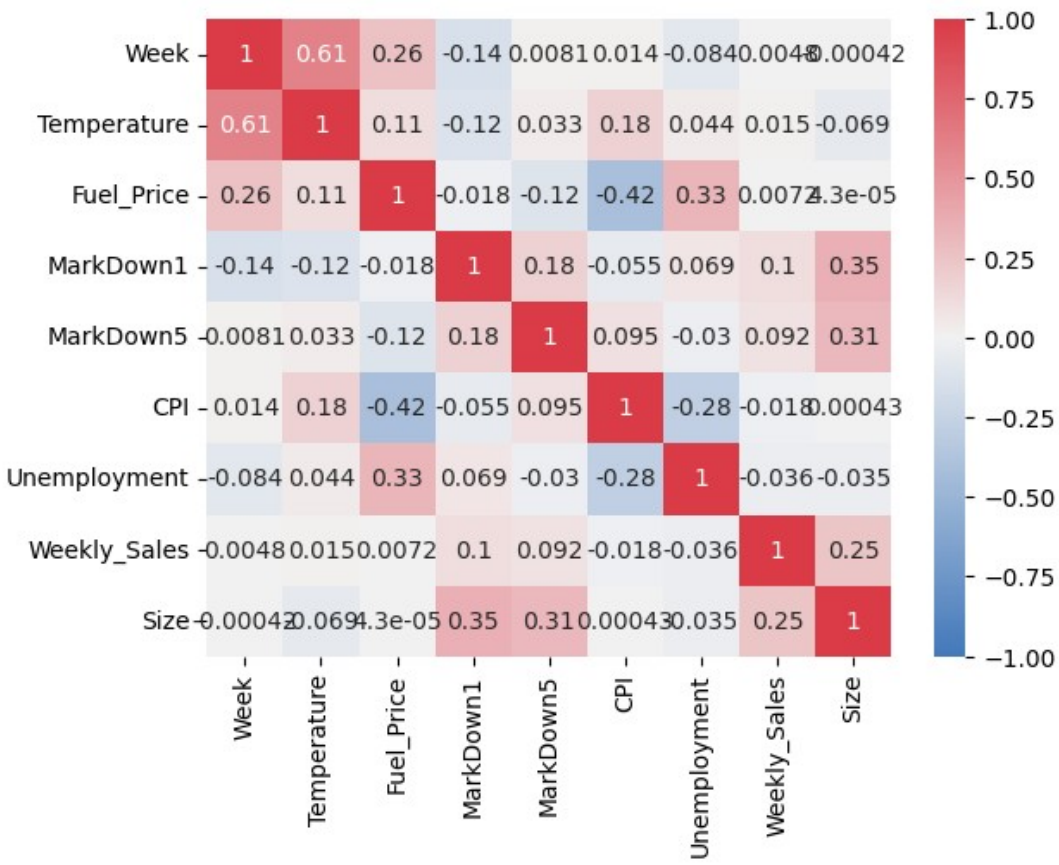

Figure 5. Correlation matrix of explanatory variables

According to Figure 5, a case of very high correlation was detected, but no cases of perfect correlation (equal to one) that could affect the model. It was also noted that several explanatory variables had little correlation with the dependent variable Weekly_Sales, and consequently, some of these variables would not be statistically significant throughout the modeling, making the prediction of the dependent variable less efficient. One of these variables could be the Week variable from level 1. Figure 6 shows the average weekly sales of the stores and their evolution over time, with only a very small variation.

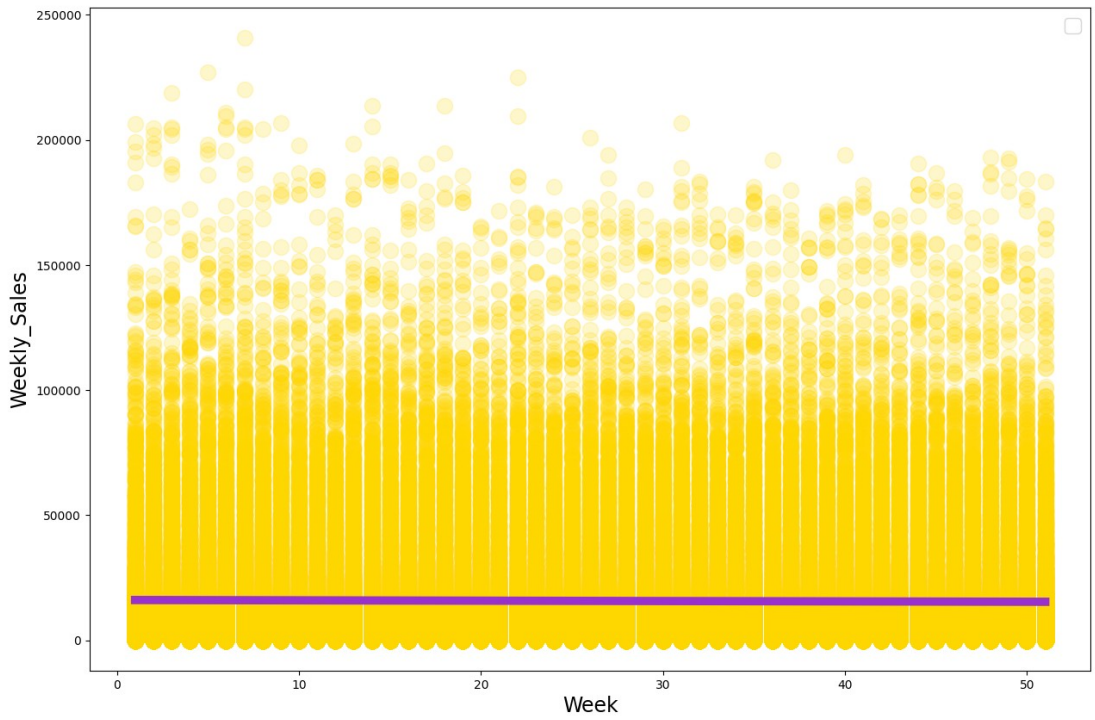

Figure 6. Average weekly sales over time

In **Figure 7**, which shows the sales of stores over the weeks, 10 stores are presented as examples. There are sales peaks during American holidays, and well-defined store groups in terms of weekly sales. At this level, a noticeable difference between the intercepts (the point from which sales start on the vertical axis) can be observed, although it is not significant, as both the intercepts and slopes appear to form small patterns. The less a store sells, the smaller the differences in intercepts and slopes.

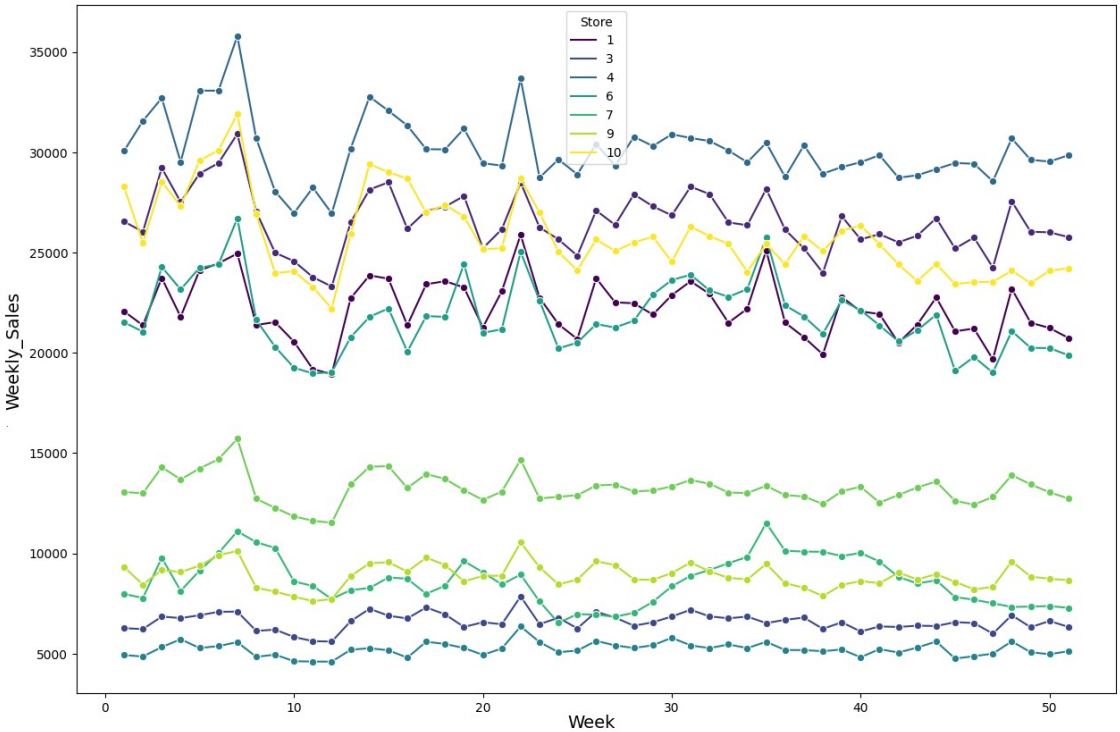

Figure 7. Weekly sales trajectories across stores

In **Figure 8**, which shows the sales of departments within a store over the weeks, 10 departments from store 1 are presented, and much more expressive differences in both intercept and slope are noted compared to the store level.

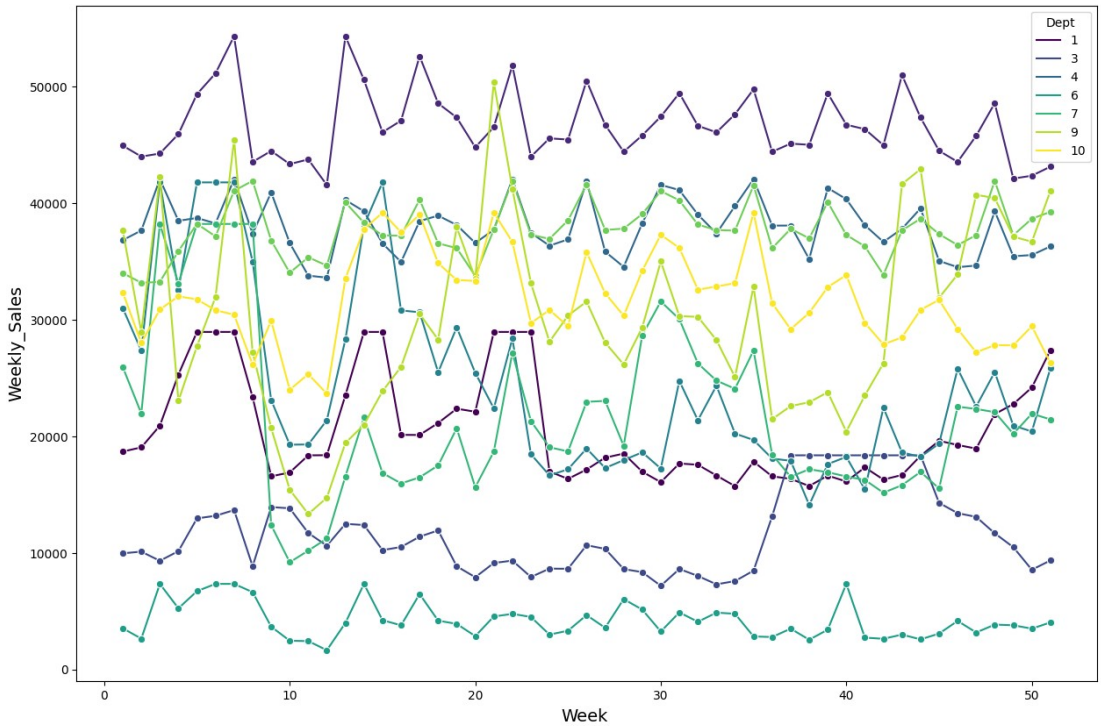

Figure 8. Weekly sales trajectories of departments within Store 1

The previous graphs and the **Size vs. Weekly_Sales** graph in **Figure 9** showed that heteroscedasticity was present. This was expected since the store size has a high correlation with weekly sales. When simulating a Linear Regression, the error terms increase roughly in proportion to the store size. The variable **Size** and the variable **Type** are W variables at the store level, and according to Hair and Fávero (2019), W variables are variables that do not vary at their respective level.

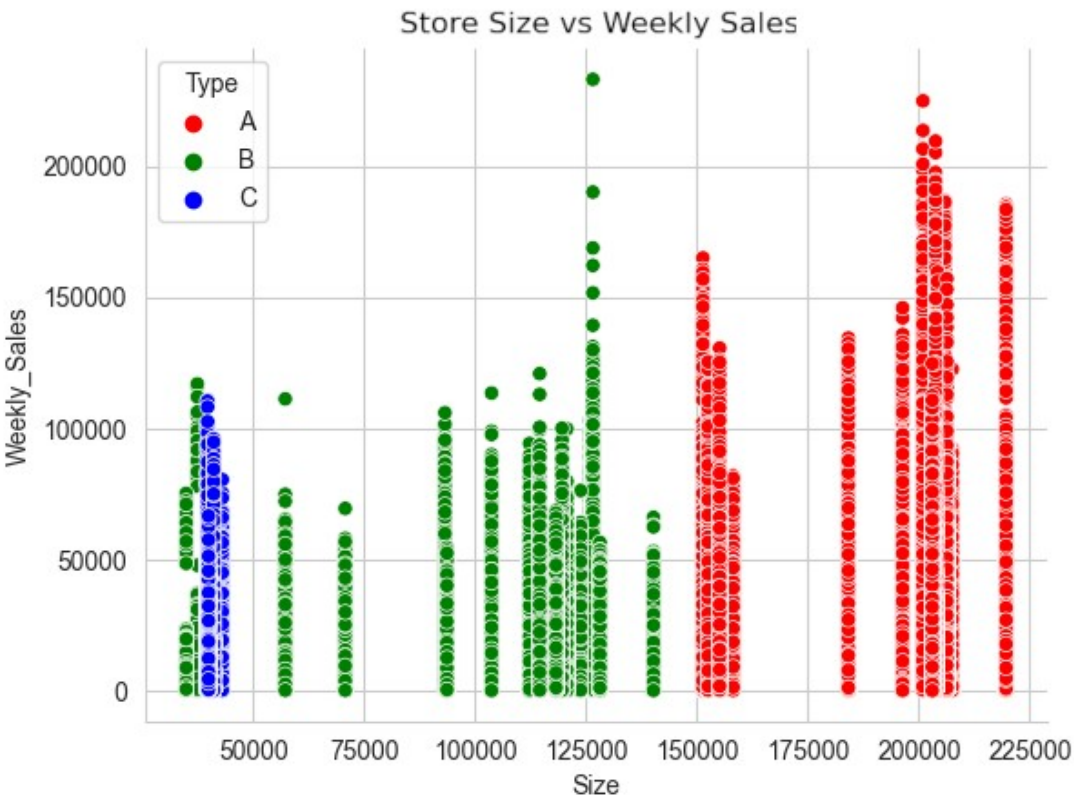

Figure 9. Store size versus weekly sales by store type

After constructing the Linear Model and then the Nonlinear Model using the "Yeo-Johnson" transformation, the **Breusch-Pagan** test should have yielded a p-value > 0.05 to not reject the null hypothesis H0 of no heteroscedasticity, but a p-value lower than 0.05 was found, confirming the presence of heteroscedasticity. The non-normality of the residuals (also called non-adherence to normality) after the **Shapiro-Francia** test, which also should have yielded a p-value > 0.05, was much lower, proving that the data needed some form of transformation before the modeling itself. Since heteroscedasticity and non-normality of

residuals persisted after the transformations, it was determined that the Nonlinear Multilevel Model would be more appropriate, as the Yeo-Johnson lambda was 0.2129547 and not 1, indicating a nonlinear model. The dependent variable resembles a Negative Binomial distribution.

Thus, the first step was to develop null models for Linear Regression, Nonlinear Regression with "Yeo-Johnson" transformation, Negative Binomial, and Multilevel Negative Binomial to compare their statistical measures and determine which best fit the data and the distribution of the dependent variable.

To make the HLM3 nonlinear, the column **Weekly_Sales**, transformed by the "Yeo-Johnson" lambda, was used as the dependent variable.

Model Selection

When estimating the null models and comparing their respective LogLik values, it was observed that the **Multilevel Negative Binomial Model**, which would have been the best model option according to the frequency graph of the dependent variable, did not fit the data well, as it had a very high LogLik value in absolute terms. Therefore, **GLM models** were estimated to later compare them with other multilevel models. The LogLik values for all of them were higher in absolute terms than the LogLik of the previous model, except for the Nonlinear Model with the “Yeo-Johnson” transformation, as shown in **Figure 10**. It was concluded that the ideal model would be the **Nonlinear Hierarchical Model with “Yeo-Johnson” transformation**. Other statistical metrics like **AIC** and **BIC** were used, and they were similar to LogLik in terms of selecting the best model.

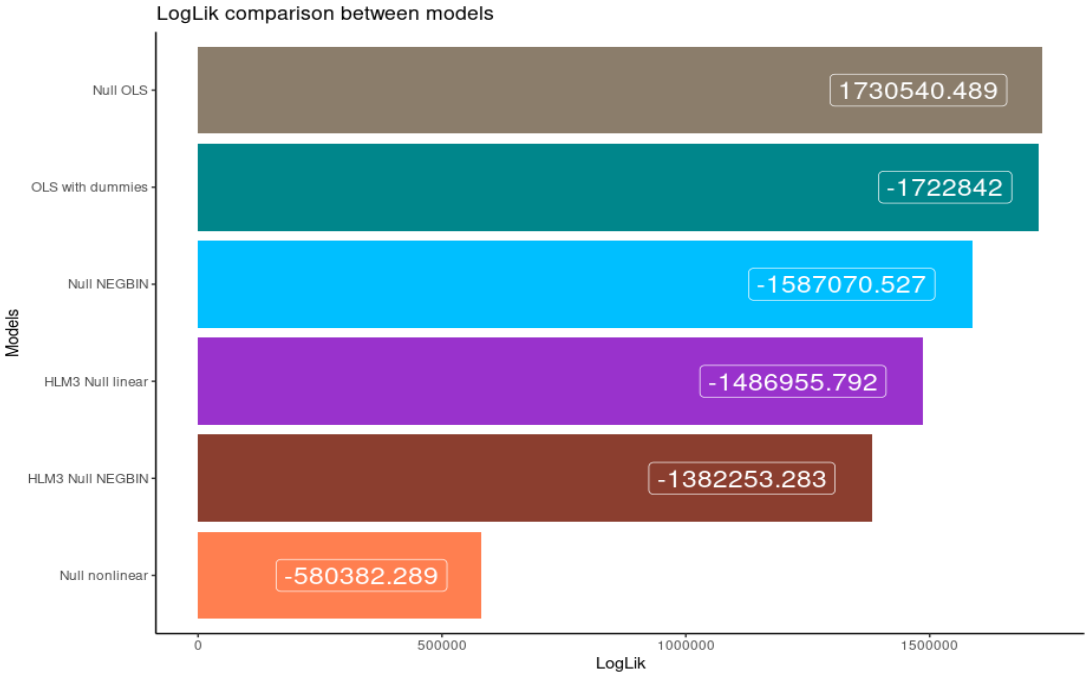

Figure 10. Model comparison based on goodness-of-fit metrics

Figure 11 shows that the "Yeo-Johnson" transformation greatly improved the data distribution. It somewhat normalized the data, which resembled a negative binomial distribution, but not enough to achieve homoscedasticity or make the distribution adhere to normality.

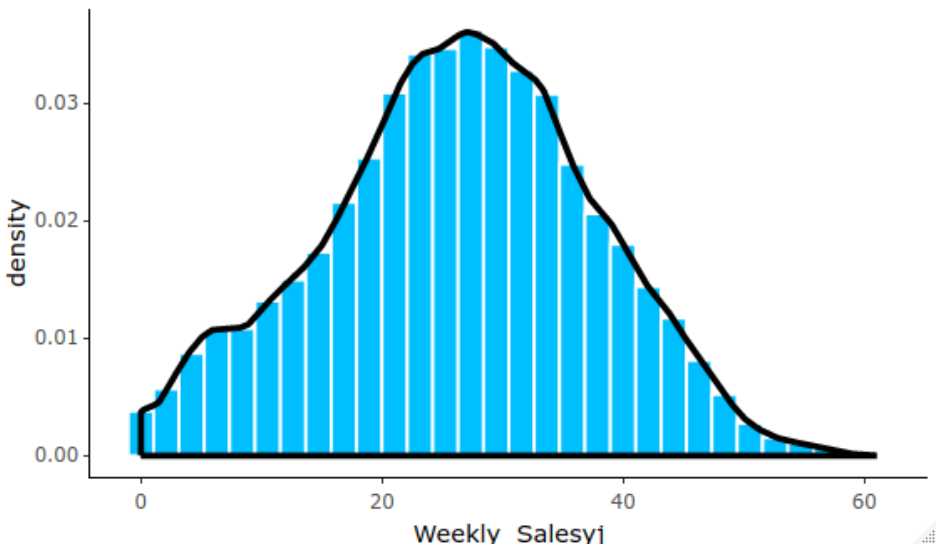

Figure 11. Distribution of the dependent variable after Yeo–Johnson transformation

After choosing the **Nonlinear Hierarchical Multilevel Model HLM3**, the next step was to calculate the **ICC's**. The formula for the linear HLM3 null model is shown below in equations 3, 4, and 5.

$$Weekly_Sales_{tjk} = \beta_{0jk} + \varepsilon_{tjk} \tag{3}$$

$$\beta_{0jk} = \delta_{000} + \tau_{00k} + v_{0jk} \tag{4}$$

$$Weekly_Sales_{tjk} = \delta_{000} + v_{0jk} + \tau_{00k} + \varepsilon_{tjk} \tag{5}$$

Since a **Nonlinear HLM3** model was used, the formula after the "Yeo-Johnson" transformation is shown below in equation 6.

$$\frac{Weekly_Sales_{tjk}^{0.2129547} - 1}{0.2129547} = \delta_{000} + v_{0jk} + \tau_{00k} + \varepsilon_{tjk} \tag{6}$$

where **β0** is the intercept with random effects; **δ00** is the overall intercept; **v0jk** and **τ00k** are random intercept values; and **εtjk** is the error.

In equation 7, the **Weekly_Sales** variable was transformed into **Weekly_Salesyj** until the end of the modeling process, where the inverse transformation is applied to the actual predicted values.
Next, the **R code** for the null model is in equation 8.

$$Weekly_Salesyj_{tjk} = \delta_{000} + v_{0jk} + \tau_{00k} + \varepsilon_{tjk} \tag{7}$$

$$\begin{aligned} model_{nullyj} = lme(\text{fixed} = Weekly_Salesyj \sim 1, \\ random = list(Store = \sim 1, Dept = \sim 1), \\ data = data, method = \text{REML}) \end{aligned} \tag{8}$$

where **lme** is one of the functions in R used for multilevel modeling; **fixed** is the field for inputting the dependent variable as a function of the fixed variables; **random** is the field for inputting the layers of the multilevel model; **data** is the field for inputting the dataset; and **method** is the field for inputting the estimation method.

In the null model, the dependent variable **Weekly_Salesyj** is a function of the intercept (i.e., 1). The “fixed” field is the fixed part of the model, where other model variables are placed. **Restricted Maximum Likelihood** (REML) refers to the restricted maximum likelihood estimation method used to estimate the parameters of the linear mixed model, making the processing less demanding, and “random” is the part of the formula for random values, where the layers are placed as a function of the variable of layer 1, **Week**.

It is important to clarify that, in this study, random effects were only allowed for **Week**, which was set as a function of the layers in "random".

After running a function to discover the random values of the null model, the output values were shown in **Table 5**. These values must be used for the ICC calculation.

Table 5. Random effects estimates of the nonlinear HLM3 null model

	Components	Estimatives	Std_Err	z	p.value
1	Var(t00k)	19.809297	4.64915116	4.260842	0
2	Var(v0jk)	129.688743	3.25323939	39.864494	0
3	Var(e)	5.044328	0.01853225	39.864494	0

Based on **Table 5**, the **ICC’s** (Intraclass Correlation Coefficients) for the time, store, and department levels were calculated, and all were statistically significant. According to **Coelho (2011)**, the ICC is the ratio of the contextual-level variance to the sum of the individual and contextual-level variances. In this table, the confirmation of what was seen in the previous figures was observed: the random intercept value **v0jk** indicates that there are differences in the sales intercepts between departments, a very significant difference as seen in **Figure 8**. The random intercept value **t00k** indicates that there are differences in the sales intercepts between departments of different stores over time, and the **e** value shows that time has little influence on weekly sales, as they change very little over time and are only accentuated during holidays. In **Equations 9, 10, and 11**, the ICC’s for the department, store, and time levels were calculated, respectively.

$$ICC_{departamento} = 129.688743 / (129.688743 + 19.809297 + 5.044328) = 0{,}839179215 \quad (9)$$

$$ICC_{loja} = 19.809297 / (129.688743 + 19.809297 + 5.044328) = 0{,}128180364 \quad (10)$$

$$ICC_{tempo} = 1 - 0{,}128180364 - 0{,}839179215 = 0{,}032640421 \quad (11)$$

Conclusions: **83.92%** of weekly sales are caused by the department level, **12.82%** by the store level, and **3.26%** by the time level.

The ICC calculation is so efficient that it reveals important information that other models could not. The correlation map had already revealed that all time and store variables did not correlate well with weekly sales, except for the **Size** variable. After calculating the ICC's, The results indicate that most of the variance in weekly sales occurs at the department level, indicating that differences between departments within the same store are more relevant for explaining variation in sales than differences between stores. If more department-level variables were included in the dataset, the predictive power of the model would become increasingly efficient.

Since the influence of time was weak, the influence of the same layers on weekly sales during holidays was calculated to show the efficiency of the ICC's. In **Equations 12, 13, and 14**, the ICC's for the department, store, and time levels were calculated using the new data shown in **Table 6**.

Table 6. Random effects estimates of the nonlinear HLM3 null model during holidays

	Components	Estimatives	Std_Err	z	p.value
1	Var(t00k)	19.36586	4.323561	4.479146	0
2	Var(v0jk)	29.01431	1.335189	21.730488	0
3	Var(e)	83.72326	1.253353	66.799434	0

$$ICC_{departament} = 29.01431 / (29.01431 + 19.36586 + 83.72326) = 0.219633283 \quad (12)$$

$$ICC_{store} = 19.36586 / (29.01431 + 19.36586 + 83.72326) = 0.146596194 \quad (13)$$

$$ICC_{time} = 1 - 0.146596194 - 0.219633283 = 0.633770523 \quad (14)$$

During holidays, **21.96%** of sales is due to the department level, **14.66%** to the store level, and **63.38%** to the time level, thus proving the effectiveness of the ICC's. The influence of the time level on sales would increase if only data where the variable **IsHoliday** equals 1

(i.e., holidays) were used. The sales peaks during holidays indicate that both the intercepts and the sales growth rate increase over time.

Illustrative Example: Fuel Price as Response Variable

An additional illustrative example is presented, transforming the explanatory variable **Fuel_Price** into a dependent variable. **Fuel_Price** was chosen To illustrate that time would become highly significant in some cases but also to show another situation.

When applying the function to find the random values of this other null model, the respective outputs were shown in **Table 7**, and in **Equations 15, 16, and 17**, the ICC's for the department, store, and time levels were calculated:

Table 7. Random effects estimates of the nonlinear HLM3 null model with Fuel_Price as response

	Components	Estimatives	Std_Err	z	p.value
1	Var(t00k)	2.783165e-02	6.071580e-03	4.583922	0
2	Var(v0jk)	4.927861e-11	1.038457e-10	0.474537	0.635
3	Var(e)	4.738909e-02	1.722303e-04	275.149608	0

$$ICC_{departament} = 0 \tag{15}$$

$$ICC_{store} = 0.02783165 / (0.02783165 + 0.04738909) = 0.369999684 \tag{16}$$

$$ICC_{time} = 1 - 0{,}369999684 = 0.630000316 \tag{17}$$

When the dependent variable is **Fuel_Price**, the layer that most influences weekly sales is **time**, remembering that the variable **Fuel_Price** has a high correlation with **Week**. The random intercept component value for the department **v0jk** in the null model is almost zero, and it is not statistically significant, meaning that there are no differences in the intercept between departments concerning fuel prices. This is because it does not vary between departments in the same store weekly, only between stores and over weeks.

Consequently, the model becomes a **2-Level Nonlinear Hierarchical Multilevel Model (HLM2)** with repeated measures, where the levels are **Time** and **Store**.

Figure 12 shows the comparison of ICCs across the three examples.

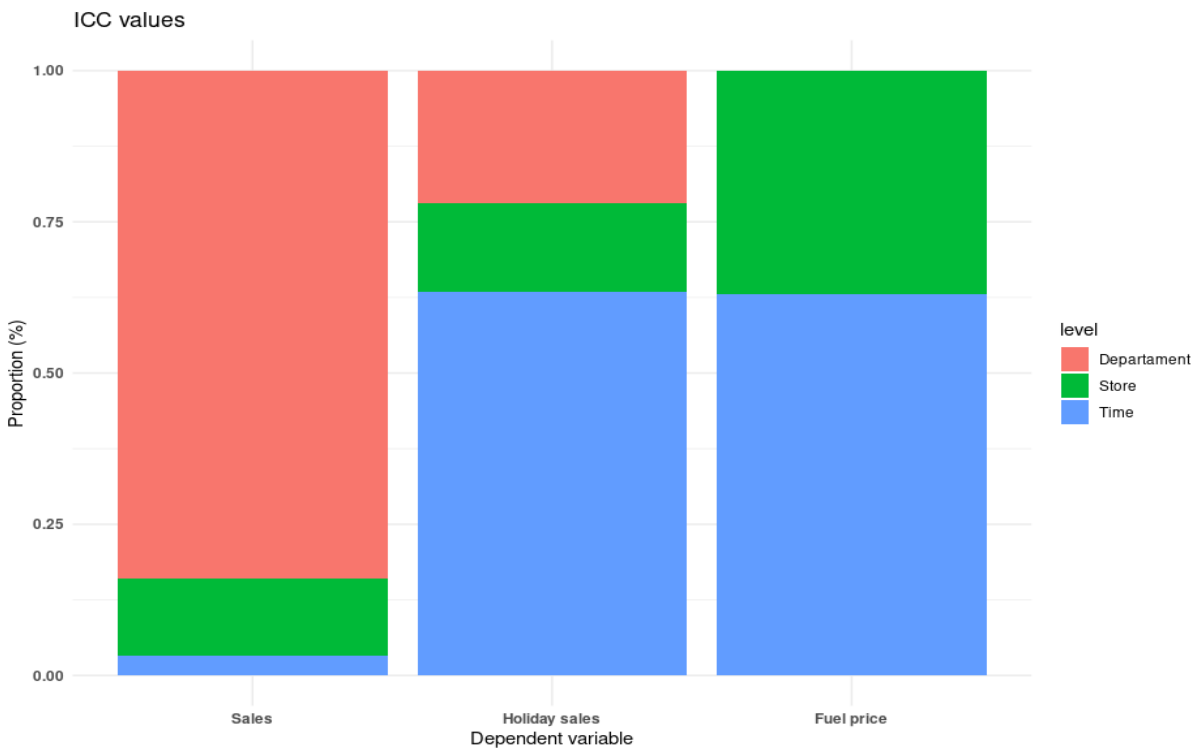

Figure 12. Comparison of ICCs across HLM3 null models

Random Intercepts and Slopes Model

After defining which model to use and calculating the ICCs, the next step is to build the chosen model, following a **Step-Up** strategy. After the null model, the next stage is to build the random intercepts and slopes model.

In the **random intercepts and slopes model**, the input data variables are included in the null model, as shown in **Formulas 18, 19, 20, 21, and 22**. **Formula 23** is the command used in R.

$$Weekly_Sales_{tjk} = \beta_{0jk} + \beta_{1jk}.Week + \beta_{tjk}.Temperature + \beta_{tjk}.Fuel_Price + \beta_{tjk}.MarkDown1 + \beta_{tjk}.IsHoliday_True + \beta_{tjk}.MarkDown5 + \beta_{tjk}.Unemployment + \beta_{tjk}.CPI + \varepsilon_{tjk} \quad (18)$$

¿

$$\beta_{0jk} = \delta_{000} + \tau_{00k} + v_{0jk} \tag{19}$$

$$\beta_{1jk} = \delta_{100} + \tau_{10k} + v_{1jk} \tag{20}$$

$$Weekly_Salesyj_{tjk} = \frac{Weekly_Sales_{tjk}^{0.2129547} - 1}{0.2129547} \tag{21}$$

$$\begin{aligned} Weekly_Salesyj_{tjk} = {} & \delta_{000} + \delta_{100}.Week + v_{0jk} + v_{1jk}.Week + \tau_{00k} + \\ & \tau_{10k}.Week + \beta_{tjk}.Temperature + \beta_{tjk}.Fuel_Price + \\ & \beta_{tjk}.MarkDown1 + \beta_{tjk}.IsHoliday_True + \beta_{tjk}.MarkDown5 + \\ & \beta_{tjk}.Unemployment + \beta_{tjk}.CPI + \varepsilon_{tjk} \end{aligned} \tag{22}$$

¿

$$\begin{aligned} model_intercept_slopeyj = {} & lme(\text{fixed} = Weekly_Salesyj \sim Week + \\ & Temperature + Fuel_Price + MarkDown1 + \\ & IsHoliday_{True} + MarkDown5 + Unemployment + CPI, \\ & random = list(Store = \sim Week, Dept = \sim Week), \\ & data = data, method = \text{REML}) \end{aligned} \tag{23}$$

where **β0** is the intercept with random effects, **β1** is the slope with random effects, **βtjk** is the slope of the fixed effects variables without random effects, **δ000** is the intercept, **δ100** is the slope of the time layer variable, **v0jk** and **τ00k** are random intercept values for department and store, respectively, **v1jk** and **τ10k** are random slope values for department and store, respectively, and **εtjk** is the error term.

In the random intercepts and slopes model, the intercepts and slopes of the variables are inserted into the model, both for random and fixed variables.

In the output of the command **summary(modelo_intercept_inclinyj)**, **Week** and **CPI** were not statistically significant. Time was influencing weekly sales very little, but still, all the random effects were statistically significant, as shown in **Table 8**.

Table 8. Random effects estimates of the HLM3 model with random intercepts and slopes

	Components	Estimatives	Std_Err	z	p.value
1	Var(t00k)	4.033367e+01	9.8147237850	4.109507	0
2	Var(t10k)	5.855669e-04	1.038457e-10	3.919336	0
3	Var(v0jk)	1.274618e+02	3.2465302037	39.260933	0
4	Var(v1jk)	4.361296e-03	0.0001342754	32.480224	0
5	Var(e)	6.346126e+00	0.0235780657	269.153786	0

In **Equation 22** and **Table 8**, it can be observed that there are two additional random effects: **v1jk** (random slope effect at the department level) and **t10k** (random slope effect at the store level), which respectively mean: “Are there differences in sales rates between departments over time?” and “Are there differences in sales rates between departments of different stores over time?” Since the results were significant, the answer to both questions is yes.

Complete Model

In the complete model, **W variables** are included. As previously mentioned, these are **level variables**, as they only vary between elements of the level. For example, in this study, the **W variables** are **Size** (store size) and **Type** (store types) from the store layer. Each store has its own type and size, which are invariant over time.

Since there is no **W variable** in the second level, "department," only the third-level formula undergoes modifications:

$$\gamma_{00k} = \delta_{000} + \delta_{001}.\text{Size} + \delta_{002}.Type_B + \delta_{003}.Type_C + \tau_{00k} \tag{24}$$

$$\gamma_{10k} = \delta_{100} + \delta_{101}.\text{Size} + \delta_{102}.Type_B + \delta_{103}.Type_C + \tau_{10k} \tag{25}$$

R Script:

$$\begin{aligned} &full_model = lme(\text{fixed} = Weekly_Salesyj \sim Week + Temperature + \\ &Fuel_Price + MarkDown1 + IsHoliday_True + MarkDown5 + Unemployment + \\ &CPI + Week:\text{Size} + Week:Type_B + Week:Type_C + \text{Size} + Type_B + Type_C, \\ &random = list(Store = \sim Week, Dept = \sim Week), \\ &data = data, method = \text{REML}) \end{aligned} \tag{26}$$

where **Type_B**, **Type_C**, and **Size** are the **W variables** from the store level.

By running the command **summary(full_model)** in R, the coefficients are displayed in the output, as shown in **Figure 13**, along with the final equation.

```
            StdDev      Corr
(Intercept) 11.25319090 (Intr)
Week         0.05125953 0.062
Residual     2.10440648

Fixed effects:  Weekly_Salesyj ~ Unemployment + CPI + Fuel_Price + MarkDown1 +      Mar
kDown5 + IsHoliday_True + Temperature + Week + Week:Size +      Week:Type_B + Week:Type
_C + Size + Type_B + Type_C
                   Value Std.Error     DF  t-value p-value
(Intercept)    23.741786 0.6705702 148171 35.40537  0.0000
Unemployment    1.296713 0.0843681 148171 15.36970  0.0000
CPI             0.160920 0.3390713 148171  0.47459  0.6351
Fuel_Price      0.041510 0.0084712 148171  4.90012  0.0000
MarkDown1      -0.046658 0.0060371 148171 -7.72856  0.0000
MarkDown5       0.052054 0.0063564 148171  8.18923  0.0000
IsHoliday_True  0.284764 0.0208569 148171 13.65321  0.0000
Temperature     0.043982 0.0104086 148171  4.22558  0.0000
Week           -0.000332 0.0035947 148171 -0.09225  0.9265
Size            4.511479 0.5964620     41  7.56373  0.0000
Type_B          2.449385 1.1039298     41  2.21879  0.0321
Type_C         -0.693303 1.7867961     41 -0.38801  0.7000
Week:Size      -0.007060 0.0030416 148171 -2.32133  0.0203
Week:Type_B    -0.009990 0.0056095 148171 -1.78087  0.0749
Week:Type_C     0.003471 0.0090915 148171  0.38180  0.7026
```

Figure 13. Estimated fixed-effect coefficients and statistical significance (final model)

The **Week** variable from the time level was not statistically significant when the full model was constructed, but it was observed that the interaction with **Size** was significant. **Size** is the variable most correlated with **Weekly_Sales**, as weekly sales increase proportionally with the size of the stores. This is why time remains significant in the first layer.

The final model is presented below. **Equation 26** is the final equation obtained based on the **Yeo–Johnson transformation**, and **Equation 27** presents the final equation of the model without the transformation.

$$Weekly_Salesyj_{tjk} = 23.741786 + 0.043982 . Temperature + 0.041510 . Fuel_Price - 0.046658 . MarkDown1 + 0.284764 . IsHoliday_True + 0.052054 . MarkDown5 + 1.296713 . Unemployment + 4.511479 . \text{Size} + 2.449385 . Type_B - 0.007060 . Week . \text{Size} - 0.009990 . Week . Type_B + v_{1jk} . Week + v_{0jk} + \tau_{00k} + \tau_{10k} . Week + \varepsilon_{tjk} \quad (27)$$

$$Weekly_Sales_{tjk} = \left(\left(Weekly_Salesyj.0.2129547\right)+1\right)^{1/0.2129547} \quad (28)$$

To verify if the differences between the null model, the random intercepts and slopes model, and the full model are statistically significant, the **lrtest()** function was used. All differences between the models were statistically significant.

When comparing the **LogLik** values from the random intercepts model and the full model with other values, the multilevel model indeed proves superior. Even the null nonlinear HLM3 model was superior to the others. It is worth remembering that part one of the comparison in **Figure 10** shows that all other models were inferior to the null nonlinear model.

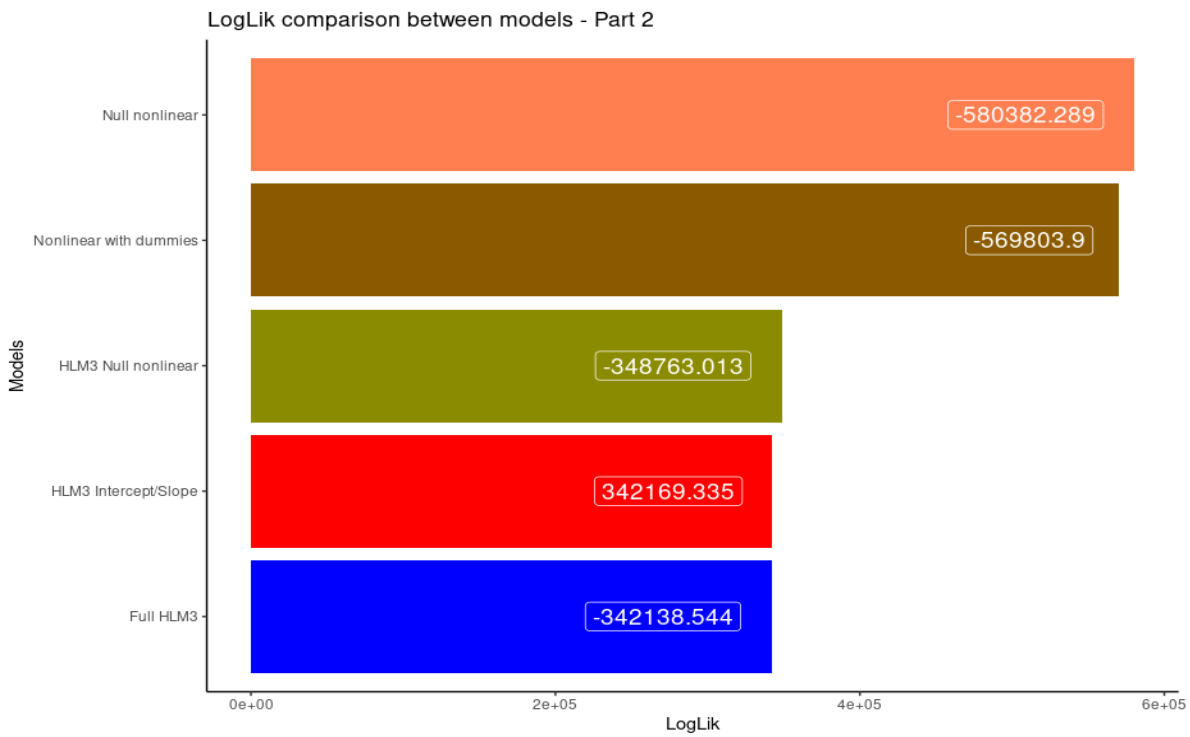

Figure 14. Final comparison of fitted models

Figure 15 shows the other statistical measures adopted throughout the model development: **AIC** and **BIC**.

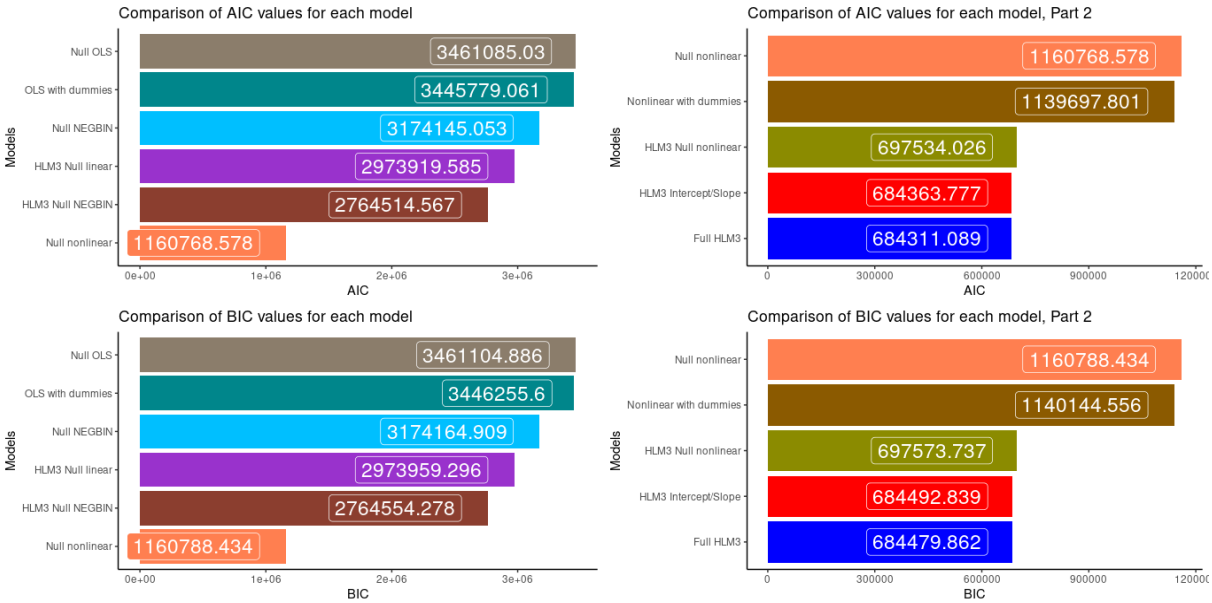

Figure 15. AIC and BIC comparison across models

Figure 16 provides an overview of all the random effects values **v0jk/v1jk** at the department level and **τ0jk/τ1jk** at the store level.

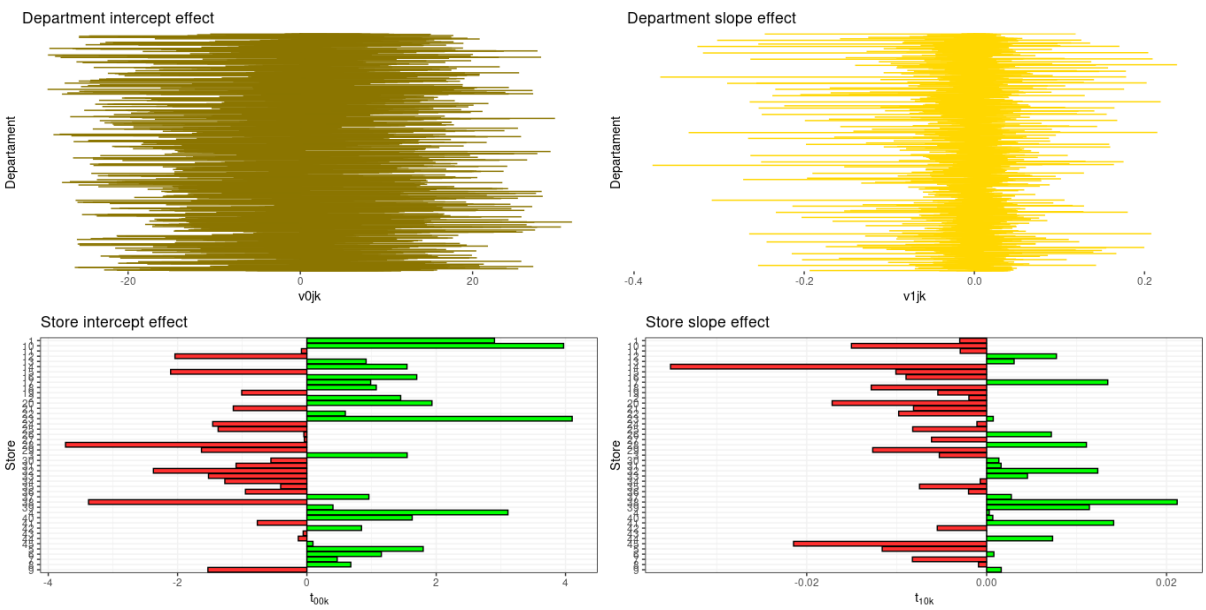

Figure 16. Overview of estimated random effects

In **Table 9**, the actual values are in the "Weekly_Sales" column, and the predicted values are in the "Dept" column (level 2). The **"FittedyjOLS"** column contains the predicted values using **linear regression**.

Table 9. Selected observed and predicted values

Index	Fixed	Store	Dept	FittedOLS	Weekly_Sales
41	5448.604	8120.840	19061.03	11076.53	17330
42	5459.716	8132.560	19049.42	11002.64	16286
43	5355.014	7985.758	18728.13	11021.83	16680
44	5702.507	8457.120	19611.26	11047.02	18322
45	5408.288	8051.769	18788.77	10956.82	19616
46	5438.132	8089.104	18827.26	10904.09	19251
47	5443.797	8093.370	18801.36	10955.67	18947
48	5175.697	7723.120	18047.29	10700.92	21904
49	5206.259	7761.625	18089.19	10645.00	22764
50	5193.127	7740.283	18014.58	10681.45	24185

The correlations of **Weekly_Sales** with almost all explanatory variables are very weak, as previously shown in **Figure 5**. The predicted values could be more accurate, but even so, the multilevel modeling managed to predict them well, even when allowing random effects only for the **Week** variable. If **linear regression** is used, with or without transformation, the prediction becomes so compromised that the **R-squared** is lower than 0.2, and only increases slightly if the dependent variable is changed to store weekly sales, thus deleting the **Dept** (department) column.

Conclusions

The use of **Multilevel Modeling** proved to be very efficient when applied to the spreadsheet data, which contains a hierarchical structure. The **ICC** values accurately showed the contribution of each level to the final model result. This algorithm captures the data well, even in a scenario of heteroscedasticity and weak correlations between explanatory and dependent variables. This efficiency is mainly due to the model's ability to calculate random effects for each level — a feature not possible with other conventional statistical models.

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